



Managing Performance and Measurement (MPM)

MPM Overview

Required PA Information

Intent

Manages performance using measurement and analysis to achieve business objectives.

Value

Maximizes business return on investment by focusing management and improvement efforts on cost, schedule, and quality performance.

Additional Required PA Information

Managing and optimizing performance helps to:

- Ensure that benefits and business performance are the leading factors in driving performance and improvement
- Change the paradigm from “process improvement leads to performance improvement” to “performance is the primary driver of process improvement”
- Use the results of measurement and analysis to manage and control performance at various work and business levels

Performance and measurement management includes:

- Setting objectives for:
 - The business
 - Measurement and performance
 - Quality and process performance
- Allocating and tracing objectives to subordinate levels in the business and processes
- Defining measurements to improve the understanding of progress towards achieving the objectives
- Analyzing measurement and performance data to:
 - Understand the relationship and interactions between performance and process
 - Define and take actions to address any observed issues with achieving objectives
 - Make the performance results and related benefits clearly visible to all stakeholders
 - Continually target and optimize performance

Measurement and performance objectives are quantitative objectives that do not require the additional rigor of statistical techniques.

Quality and Process Performance Objectives (QPPOs) apply to High Maturity activities using statistical and other quantitative techniques. These objectives include the use of statistical and other quantitative techniques on the related data.

Explanatory PA Information

Practice Summary

Level 1

- MPM 1.1 Collect measures and record performance.
- MPM 1.2 Identify and address performance issues.

Level 2

- MPM 2.1 Derive and record measurement and performance objectives from selected business needs and objectives and keep them updated.
- MPM 2.2 Develop, keep updated, and use operational definitions for measures.
- MPM 2.3 Obtain specified measurement data according to the operational definitions.
- MPM 2.4 Analyze performance and measurement data according to the operational definitions.
- MPM 2.5 Store measurement data, measurement specifications, and analysis results according to the operational definitions.
- MPM 2.6 Take actions to address identified issues with meeting measurement and performance objectives.

Level 3

- MPM 3.1 Develop, keep updated, and use the organization's measurement and performance objectives traceable to business objectives.
- MPM 3.2 Follow organizational processes and standards to develop and use operational definitions for measures and keep them updated.
- MPM 3.3 Develop, keep updated, and follow a data quality process.
- MPM 3.4 Develop, keep updated, and use the organization's measurement repository.
- MPM 3.5 Analyze organizational performance using measurement and performance data to determine and address performance improvement needs.
- MPM 3.6 Periodically communicate performance results to the organization.

Level 4

- MPM 4.1 Use statistical and other quantitative techniques to develop, keep updated, and communicate quality and process performance objectives that are traceable to business objectives.
- MPM 4.2 Select measures and analytic techniques to quantitatively manage performance to achieve quality and process performance objectives.

- MPM 4.3 Use statistical and other quantitative techniques to develop and analyze process performance baselines and keep them updated.
- MPM 4.4 Use statistical and other quantitative techniques to develop and analyze process performance models and keep them updated.
- MPM 4.5 Use statistical and other quantitative techniques to determine or predict achievement of quality and process performance objectives.

Level 5

- MPM 5.1 Use statistical and other quantitative techniques to ensure that business objectives are aligned with business strategy to optimize performance.
- MPM 5.2 Analyze performance data using statistical and other quantitative techniques to determine the organization's ability to satisfy selected business objectives and identify potential areas for optimizing performance.
- MPM 5.3 Select and implement improvement proposals based on the statistical and quantitative analysis of the expected effect of proposed improvements on meeting and optimizing business, quality, and process performance objectives.

Additional PA Explanatory Information

The term "process performance" refers to the "working level" where the solutions are being developed or delivered, while the term "business performance" refers to the business or organizational level. "Performance" can refer to either or both levels. For example, collect measurement and performance data at the "work level" and aggregate data to enable organizational performance analysis at the business level.

A robust measurement program includes and addresses the following types of objectives. [Table MPM-1: Progression and Relationship of Objectives](#) shows the progression and differences of the types of objectives. In each case, these objectives should be traceable to each other.

Table MPM-1: Progression and Relationship of Objectives

Objective	Explanation
Business Objective: Achieve 33% faster delivery per release.	Organization realizes that to increase their competitive position, they need to deliver releases at a faster rate. Results from objective evaluations indicate there is a significant amount of time resulting from ineffective manual testing, which is contributing to the current release velocity.
Measurement and Performance Objective: Increase automated testing to 30% code coverage per release.	The amount of time spent developing and executing manual test scripts is a key contributor to the length of the testing process. An improvement proposal has been identified to convert some of the manual testing to automated testing.
Quality and Process Performance Objective (QPPO):	The organization has implemented a performance improvement by converting some of the manual

Objective	Explanation
Achieve average initial pass rate of 86%, +/- 10% for automated testing per Sprint.	testing to automatic testing. By applying statistical process control to the automated testing, the organization has been able to significantly reduce the length of the testing process. Achieving this QPPO on a per Sprint basis enables the organization to maintain this level of performance for releases.

In the above examples, the business objectives and measurement and performance objectives are indicative of processes that meet the intent and value of practices in Practice Groups 1, 2, and 3. The QPPO example shows the evolution to achieving Practice Groups 4 and 5.

Related Practice Areas

Causal Analysis and Resolution (CAR)

Estimating (EST)

Monitor and Control (MC)

Context Specific

Agile Development

Context Tag:	Agile Development
Context:	Integrate agile techniques and ceremonies with other processes.

[Table MPM-2: Example Agile Measurement Activities](#) provides examples of where measurement and performance play a role in an agile project.

Table MPM-2: Example Agile Measurement Activities

Agile Activity or Ceremony	Example Measurements
Release Planning	Assignment of backlog items to future Sprints
Backlog Grooming	Epics and story points remaining
Sprint Planning	Planned versus actual Sprint velocity
Sprint Review/Demo	Sprint velocity (story points completed per Sprint), burn down chart
Retrospective	Average Sprint velocity versus actual Sprint velocity

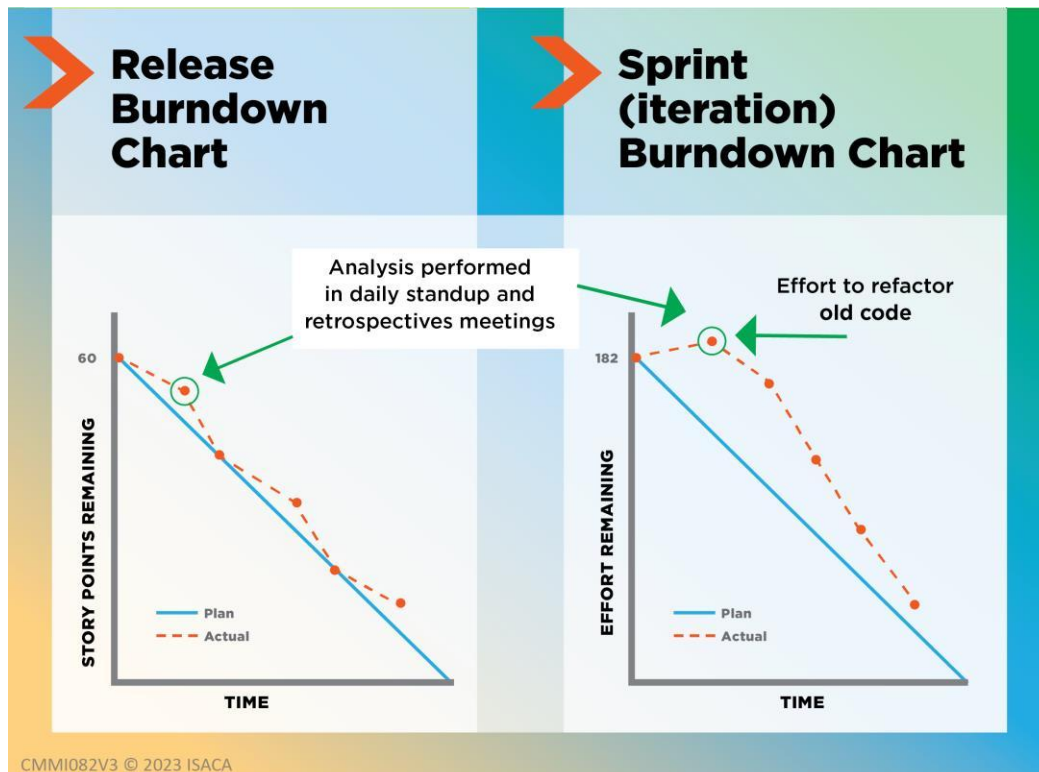
An agile development project can improve performance by:

- Defining objectives that go beyond a single iteration. Objectives might include achieving a certain velocity over successive Sprints to be able to satisfy the business need.
- Recording clear and usable operational definitions of measures
- Defining how velocity is calculated:
 - Identified the canonical value of 1, i.e., team agrees on the operational definition of what one story point is

- Agreed to scale to be used for story points, most commonly using the Fibonacci sequence
- Identifying and assigning story points to Sprints
- Application of average team Sprint velocity to future Sprints

Typically, agile teams use burndown charts for releases and Sprints to monitor progress and help assess performance.

Figure MPM-1: Release Burndown Chart & Sprint (Iteration) Burndown Chart



The Release Burndown Chart in [Figure MPM-1: Release Burndown Chart & Sprint \(Iteration\) Burndown Chart](#) shows the number of story points remaining over time, tracked within each Sprint, and representing all of the work for a release consisting of several Sprints. The performance objective of this work is to burn down a target number of story points within a release within a given timeframe using a set of resources. This chart shows:

- The number of story points completed ("Value delivered")
- Those points forecasted across all planned Sprints
- All of the planned work for a release

The performance objective of this work is related to the value delivered, and used to help forecast what might be delivered in future Sprints related to the release.

Measurements are used to track the remaining story points, and the performance analysis determines any deviation and the associated reason. For example, overcommitting to story points, or people unexpectedly on sick leave. Actions may be taken, such as postponing user stories to later releases, adding more resources, or removing impediments.

The Sprint (Iteration) Burndown Chart in [Figure MPM-1: Release Burndown Chart & Sprint \(Iteration\) Burndown Chart](#) shows the remaining story points that have been forecasted for the Sprint, which are updated continuously. The performance objective is to complete the planned stories within the Sprint using a consistent set of resources. The measurement gives the remaining points to be delivered, and the performance analysis collected through visual information indicators and daily standups informs the team of any deviation and their reasons, e.g., the code is difficult to modify. Appropriate actions are taken to remove impediments.

Data

Context Tag:	CMMI-DATA
Context:	Use processes to incorporate data management best practices as an integral part of the solution.

Identify and leverage measurements to assess the achievement of the objectives for managing data. Consider a comprehensive approach, across all business units, business processes, projects, and applications, for measuring and reporting on data quality dimensions that are important to the project or organization. Analyze the measurement data, including consideration of error rates and quality thresholds, to assess the effectiveness of the approach for managing data and take appropriate corrective action when measures indicate data is outside of tolerance levels, or when there is misalignment with business objectives.

Examples of data measures to consider include:

- Value of metadata management, e.g., link to cost containment, operating efficiency, process effectiveness
- Performance of key processes and procedures, e.g., frequency of use per attribute, number of data stores per attribute, quantity of applications per attribute
- Criticality of data attributes to applications, e.g., which are core, for what process, used in which application, included in which calculation processes
- Costs associated with the movement of data across the lifecycle, e.g., how much is at risk, especially if the lineage is fragmented
- Tracking progress toward a single authoritative source
- Quality of metadata breadth, depth, scope, availability, timeliness, accuracy, duplication, conformity, linkages, and clarity
- Metadata metrics, e.g., adoption, percent complete
- Satisfaction of Service Level Agreements (SLAs) regarding platform or technical performance and capabilities
- Number of data management breaches in comparison to the defined approach
- Percent of roles and responsibilities supporting the governance of data management

DevSecOps

Context Tag:	DevSecOps
Context:	DevSecOps is a mindset, a culture, and set of practices that fosters close cooperation between development, operations, and security to plan, develop, test, deploy, release, and keep updated a secure solution.

DevSecOps teams leverage automated tools and processes to perform their work. This enables the team to identify process objectives and to collect and analyze a variety of measures to determine progress against them. Measures are selected and used to achieve a balance between business needs, process adherence, and effectiveness. Measures are also used by the team to effectively manage their work and contribute to achievement of business objectives. It is important to consider the full context of the overall process and activities when analyzing measurement results. DevSecOps teams use a variety of automated tools to capture significant amounts of data about the software process from development through deployment and operations. When determining measures, a DevSecOps team typically considers:

- Reliability
- Security
- Stability
- Performance
- Cost
- Process Lead Time
- Ease of Use
- Continuity
- Deployment Queue Rate
- Consistency

Safety

Context Tag:	CMMI-SAF
Context:	Use processes to incorporate safety considerations as an integral part of the solution, work, project, and organization.

Every solution is likely to have some level of safety requirements included as part of its design and operation. These requirements typically include safety objectives and related measurements that are used to verify that the safety requirements have been addressed.

Security

Context Tag:	CMMI-SEC
Context:	Use processes to incorporate security considerations as an integral part of the solution, work, project, and organization.

Every solution is likely to have some level of security requirements included as part of its design and operation. These requirements typically include security objectives and related measurements that are used to verify that the security requirements have been addressed.

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

An acquirer establishes measurement objectives for its activities and work products and for supplier activities and deliverables. Measurement and analysis of solution components provided by suppliers is important to effectively manage the quality and costs of the project. Careful management of supplier agreements can provide insight into data that supports supplier performance analysis. The supplier agreement should include:

- Measures for the supplier to collect and report
- Data collection processes and timing for the supplier to use
- Expected analysis of supplier data
- Required storage of supplier data

Identify additional measures to track and achieve interface or connection interoperability in projects where multiple acquired solutions deliver a capability to the end user or where there are relationships with other projects to acquire joint capabilities.

The acquirer specifies measures to:

- Gauge its own progress and output
- Monitor supplier performance and progress per contractual requirements
- Gain insight into the status of the evolving solutions acquired

Level 1

MPM 1.1

Required Practice Information

Practice Statement

Collect measures and record performance.

Value

Enables performance management to increase likelihood of meeting objectives.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Collect available data to provide the team and management with insight into performance. Identify data that can be used to manage the project. Identifying and using this data leads to effective performance management.

If limited data are available, senior management should identify the basic set of information needed to manage the project.

The collection of similar performance measurements at the work level can be aggregated upwards to form the basis for performance management within an organization. For success, senior management and other stakeholders should record and communicate current performance needs and business objectives.

Example Activities

Example Activities	Further Explanation
Identify available measures and collection methods.	Collection methods may include: • Surveys • Observation • Direct recording • Statements by customers, clients, or other stakeholders • Industry comparison Current business performance data may include: • Customer satisfaction • Sales • Profit and loss Process performance data may include: • Adherence to schedule and budget

Example Activities	Further Explanation
	• Estimates and significant deviations • Quality data, e.g., defects, warnings, customer complaints, returns, incidents
Collect and record measures to understand performance.	Determine if collected performance data is relevant and critical to the work and the business. May include: • Results from measurement and collection • Discussion and interpretation of results • Usage of results Record data from: • Business performance • Current and planned process performance • Performance improvements Performance data may: • Include single or smaller efforts addressing performance and performance improvement • Originate from plans that have been developed at any level, e.g., performance initiative, process initiative, team, organization, product characteristics
Record performance and communicate results.	Results may be stored in available tools and reported periodically or on an event-driven basis.

Example Work Products

Example Work Products	Further Explanation
Measures	May include: • Product measures • Process measures • Quality measures • Customer measures • Employee measures
Performance analysis results	May include: • Business performance • Customer satisfaction • Employee satisfaction • Quality expectations

Context Specific

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

Acquiring organizations need to identify the data needed to provide insight into supplier activities and performance. The supplier agreement must include any data expected from suppliers.

MPM 1.2

Required Practice Information

Practice Statement

Identify and address performance issues.

Value

Improves the ability to achieve objectives and increases customer satisfaction.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Identify issues by comparing performance with available performance data such as:

- Industry data
- Customer requirements
- Previous performance
- Planned performance
- Objectives

Record performance issues, their causes, and possible solutions. Proposed suggestions can be used to spread successful performance improvements to other projects.

Example Activities

Example Activities	Further Explanation
Collect measurements and derive performance data.	
Review performance.	
Identify issues associated with performance.	

Example Activities	Further Explanation
Understand the causes for performance issues.	
Make suggestions to improve performance.	
Address performance issues based on suggestions.	

Example Work Products

Example Work Products	Further Explanation
List of performance issues	May include: • Issues characteristics • Reasons for issues • Resolution options • Actions taken for resolution
List of suggestions	May include: • Description • Suggestions selected for implementation

Level 2

MPM 2.1

Required Practice Information

Practice Statement

Derive and record measurement and performance objectives from selected business needs and objectives and keep them updated.

Value

Aligns measurement and performance activities to increase the likelihood of achieving business results.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Measurement and performance objectives, along with the related data, provide stakeholders with the information needed to understand performance against business objectives. This supports realistic planning and helps avoid duplicative and ineffective performance improvement activities. This includes addressing measurement objectives at the project level. Projects may determine that they need to establish a project measurement repository to support these activities.

Measurement objectives provide a common basis for employees and senior managers to measure progress and improve or remove ineffective practices. Objectives and measures help to:

- Provide insight into actual size and effort completed compared to the plan
- Provide insight into schedule fluctuations and progress
- Provide insight into actual costs compared to plan
- Identify unplanned work or scope creep
- Determine the cost and schedule impact of rework
- Evaluate the effectiveness of defect detection throughout the solution lifecycle
- Evaluate supplier performance

Analysis of performance information can indicate where modifications to the identified information needs and objectives are required. Determine whether the value of the result is aligned with the resources dedicated to doing the work.

Sources of information needs and objectives include:

- Customer and stakeholder expectations
- Established management objectives

- Strategic plans
- Business plans
- Formal requirements or contractual obligations
- Work plans
- Work performance monitoring
- Process improvement plans
- Interviews with senior managers and others who have information needs
- Recurring or persistent issues
- Supplier agreements and contractual requirements
- Experiences of other workgroups or organizational entities
- External industry benchmarks

Example Activities

Example Activities	Further Explanation
Identify current or planned performance improvements.	Collect information about performance improvements, e.g., intention, objectives, actions, timeframe.
Record and prioritize business needs and objectives.	Set priorities within the limits of available resources.
Review and keep updated measurement and performance objectives.	• In the review, include the purpose, value, and intended uses of objectives • Record, keep updated, and have stakeholders review measurement and performance objectives • Involve users and providers of measurement, performance, and analysis results in setting measurement objectives and deciding on plans of action
Review business needs and objectives against measurement and performance objectives, as needed, with affected stakeholders.	Identified business needs and objectives can be refined and clarified as a result of setting measurement and performance objectives. Reviewing business needs and objectives against measurement and performance objectives is an iterative process. It is possible that: • Measurement and performance objectives are not aligned with business needs and objectives • Initial descriptions of business needs are ambiguous

Example Work Products

Example Work Products	Further Explanation
Measurement and performance objectives	Examples of objectives include to: • Meet commitments for: ○ Budget ○ Schedule

Example Work Products	Further Explanation
	○ Quality • Increase: ○ Customer satisfaction ○ Employee satisfaction ○ Innovation and creativity • Reduce: ○ Customer complaints ○ Errors and rework ○ Environmental impact

Context Specific

People

Context Tag:	CMMI-PPL
Context:	Use processes to align workforce performance and business objectives.

Incorporate workforce management needs and constraints within measurement and performance objectives.

Safety

Context Tag:	CMMI-SAF
Context:	Use processes to incorporate safety considerations as an integral part of the solution, work, project, and organization.

Incorporate safety-related aspects within measurement and performance objectives.

Security

Context Tag:	CMMI-SEC
Context:	Use processes to incorporate security considerations as an integral part of the solution, work, project, and organization.

Identifying and categorizing security measurements help to quantify the potential impact and severity of the security issues, enabling more timely responsiveness for meeting security objectives. Examples of security measurement categorization may include:

Information Technology

- Number of systems, servers, and users with known or discovered security issues and vulnerabilities
- Number of expected or potential issues, threats, and vulnerabilities
- Number and frequency of access issues
- Number of Secure Sockets Layer (SSL) Certificates configured incorrectly
- Frequency of access to critical enterprise systems and solutions by third parties

- Number of days to deactivate former employee credentials
- Volume of data or transactions requiring encrypted transmission
- Volume or number of documents, records, or drawings requiring Classified, Secret, or Top Secret protection
- Frequency or periodicity of data and information back-ups
- Remote workforce security constraints, e.g., providing a secure network or Virtual Private Network (VPN)

Solution Development

- Number of solution components, subsystems, and systems with known or discovered issues and vulnerabilities
- Type of threats and vulnerabilities discovered with the solutions or solution components
- Number of security issues related to the security architecture
- Number of externally-made parts, components, or materials in the solution
- Number of external suppliers or vendors providing to the solution
- Number of third-party developers, integrators, or testers of the solution
- Number of operations, process steps, or items of equipment requiring signal suppression or shielding

Service Delivery

- Number and frequency of internal versus external service delivery security issues, threats, and vulnerabilities
- Number of service system processes, subsystems, and components with known issues and vulnerabilities
- Number and type of threats and vulnerabilities identified and discovered with the service delivery

Organizational

- Number of security evaluations planned versus conducted
- Number of security weaknesses discovered per security evaluation
- Number of threats and vulnerabilities related to security policy violations
- Effort and cost for security actions and steps, e.g., per issue, objective, or evaluation
- Number of employees who cannot receive a security clearance due to issues discovered in background checks

Supplier Management

- Number of security issues, threats, and vulnerabilities identified with the supplier's solutions or solution components
- Number of supplier work product security evaluations completed (planned versus actual)
- Number of supplier security process evaluations completed (planned versus actual)
- Service levels or timeframes for suppliers to address known security issues and for the recipient to verify resolution

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

For objectives defined in supplier agreements, the acquirer defines measures that provide insight into the performance of suppliers, including the quality of their deliverables.

Measurement objectives focus on acquirer performance, supplier performance, and understanding the effects on customer, operational, and financial performance. Supplier measurement objectives help to define and track service level requirements recorded in the supplier agreement.

Derive measurement objectives needed to:

- Maintain alignment to project objectives and provide results that keep a project on track to its successful conclusion
- Support the organization's ability to develop an infrastructure that reinforces and grows acquirer capabilities, including those related to processes, people, and technologies
- Support the ability to monitor and manage financial results and customer expectations

Review appropriate measurement objectives with potential suppliers throughout the solicitation, obtaining their feedback and commitment.

Example supplier deliverables include:

- Quality performance data
- Performance against agreed to service levels
- Cost and effort data

Include required supplier deliverables in the supplier agreement.

MPM 2.2

Required Practice Information

Practice Statement

Develop, keep updated, and use operational definitions for measures.

Value

Increases the consistency of measures and the likelihood that business needs and objectives are met efficiently and effectively.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Operational definitions of measures enable consistent collection, analysis, use, and understanding of measurements, performance, and results.

Use of operational definitions increase the:

- Data quality
- Understanding and use of measurements
- Usefulness of historical information
- Ability to make informed management decisions

State operational definitions in clear and unambiguous terms. Operational definitions help to establish a consistent understanding of measurements and their use. Operational definitions address two important criteria:

- Communication: What has been measured, how was it measured, what are the units of measure, and what has been included or excluded?
- Repeatability: Can the measurement be repeated, given the same definition, and get the same results?

Define collection and storage methods to help ensure that the right data are collected consistently and stored in the right place. Storage and retrieval procedures help to ensure that data are available and accessible by the right people for future use.

Specify the analysis procedures to:

- Ensure that the necessary data will be collected
- Ensure the quality of data
- Help minimize subjective interpretation
- Lower the risk of reaching incorrect conclusions due to inappropriate analysis techniques or lack of understanding

Refine measurement and performance objectives into precise, quantifiable measures. Ensure that traceability between the measurement and performance objectives and the related measures is clear, available, bidirectional, and kept updated.

Measures can be either base or derived. Obtain data for base measures by direct measurement. Data for derived measures typically come from combining two or more base measures. Derived measures are often expressed as ratios, composite indices, or other aggregate summary measures.

Refer to [Table MPM-3: Base and Derived Measurement Examples](#).

Table MPM-3: Base and Derived Measurement Examples

Measurement Type	Examples
Base measures	Commonly used base measures include: • Measures of size, e.g., number or count of items or activities, pages, number of requirements • Measures of effort and cost, e.g., number of person hours, headcount • Quality measures, e.g., number of defects by severity, rework amount before and after delivery • Information security measures, e.g., number of system vulnerabilities identified • Customer satisfaction survey scores
Derived measures	Examples of commonly used derived measures include: • Earned value • Productivity • Rework percentage • Peer review coverage • Test or verification coverage • Reliability, e.g., mean time to failure or error • Maintainability, e.g., system or service down time • Quality, e.g., number of defects by severity/total number of defects • Information security, e.g., percentage of system vulnerabilities mitigated • Customer satisfaction trends

There are direct relationships among information needs, measurement and performance objectives, types, or categories of measurement, base measures, and derived measures. This direct relationship is depicted using some common examples in [Table MPM-4: Example Measurement Relationships](#).

Table MPM-4: Example Measurement Relationships

Example Business Objective	Information Need	Purpose	Information Type or Category	Example Base Measures	Example Derived Measures
Shorten time to delivery. Be first to market the solution.	What is the estimated delivery time?	Provide insight into schedule fluctuations and progress.	Schedule and progress	Estimated and actual start and end dates by task	Schedule Performance Schedule estimation accuracy
Increase market share by reducing costs of solutions.	How accurate are the size and cost estimates?	Provide insight into actual size and costs compared to plan.	Size and effort	Estimated and actual effort and size	Productivity
			Effort and cost	Estimated and actual cost	Cost performance Cost variance

Example Business Objective	Information Need	Purpose	Information Type or Category	Example Base Measures	Example Derived Measures
Deliver specified functionality.	Has scope or work size grown?	Provide insight into actual size compared to plan, identify unplanned growth.	Size and stability	Requirements count	Requirements volatility Size estimation accuracy
Reduce defects in solutions delivered to the customer by 10% without affecting cost.	Where are defects being inserted and detected prior to delivery?	Evaluate the effectiveness of defect detection throughout the solution lifecycle.	Quality	Number of defects inserted and detected Solution size	Defect containment Defect density
	What is the cost of rework?	Determine the cost of correcting defects.	Cost	Number of defects inserted and detected by lifecycle phase Effort hours to correct defects Labor rates	Rework costs
Reduce information system vulnerabilities.	What is the magnitude of open system vulnerabilities?	Evaluate the effectiveness of mitigating system vulnerabilities.	Information assurance	Number of system vulnerabilities identified, and number of system vulnerabilities mitigated	Percentage of system vulnerabilities mitigated
Increase innovation in solutions.	How innovative are the solutions?	Determine the level of innovation of past, current, and future solutions.	Organizational creativity	Number of new ways to do things. Innovation can include: • Sustaining or Adjacent • Disruptive • New Market • Breakthrough	Innovation trends, percentage of adjacent, disruptive, breakthrough and new market innovations, processes that address innovations,

Example Business Objective	Information Need	Purpose	Information Type or Category	Example Base Measures	Example Derived Measures
					incentives for innovation

Example Activities

Example Activities	Further Explanation
Identify existing measures that address measurement objectives and are available or collected from current work products, processes, or transactions.	Specifications for measures, sources, and collection mechanisms may already exist.
Select measures that provide insight into the organization's quality and performance.	Select measures for the selected product and process attributes. Examples of systematic approaches to select measures include: • Goal Question Metric (GQM) • PSM (Practical Software and Systems Measurements) • AIM (Accelerated Improvement Method) Examples of criteria used to select measures include: • Relationship of measures to the organization's objectives • Coverage that measures provide over the life of the solution • Visibility that measures provide into performance • Frequency at which observations of the measure can be collected
Establish operational definitions for the selected measures.	May include: • Collection steps and rules • Functions or algorithms used to produce derived measures • Analysis procedures • Decision criteria – Numerical thresholds, targets, or limits used to determine need for action, e.g., 20% variance from plan requires replanning
Specify how to collect and store data for each required measure.	Develop data collection mechanisms and process guidance. Data collection mechanisms can include: • Manual or automated methods • Forms, templates, and tools • Mechanisms for ensuring data quality
Select data analysis methods and tools.	Issues to be considered typically include: • Choice of analysis techniques • Choice of visual display and other presentation techniques, e.g., pie charts, bar charts, histograms, radar charts, line graphs, scatter plots, tables

Example Activities	Further Explanation
	• Decisions about how to handle analysis in the presence of missing data elements • Selection of appropriate analysis tools During data analysis, consider: • Interrelationships among specified measures • Changes over time
Review operational definitions with affected stakeholders and keep them updated.	

Example Work Products

Example Work Products	Further Explanation
Operational definitions of measures	Typically includes: • Name • Description, context, and purpose • Reason for the measure, e.g., performance, legal, management • Visual display of measure (indicator) • Data elements for derived measures • Definition for each element • Data collection • How data are collected • When data are collected • Who will collect data • Forms and templates for collection • Tools for data collection • Validation and quality • Data reporting • Responsibilities • To whom data are reported • Frequency of report • Algorithm or calculation for derived measures • Any assumptions • Interpretation for the measure • Analytic techniques • Traceability to objectives and source • Cross reference to any other measures
Analysis methods and tools	
Data collection mechanisms	

Context Specific

DevSecOps

Context Tag:	DevSecOps
Context:	DevSecOps is a mindset, a culture, and set of practices that fosters close cooperation between development, operations, and security to plan, develop, test, deploy, release, and keep updated a secure solution.

The business objectives should drive the measurement selection and use. Business, organization, and project leaders must decide which measures are operationally meaningful to the organization, as well as how to implement and use them. Standard operationally defined measures used by DevSecOps teams typically include:

- **Application Change Time:** Time between code commit and production deployment. This measure includes time to build, test, and release an update. When application change times are shorter, this implies improved development efficiencies.
- **Application Deployment Frequency:** Number of deployments to production in each iteration. Lower deployment frequency might be acceptable for a mature product, while high deployment frequency is common for new or less mature products.
- **Availability:** Application uptime or downtime in a period. This is an important metric usually linked to the Service Level Agreement (SLA).
- **Change Failure Rate:** Number or percentage of failed production deployments that result in an aborted deployment or rollback. A high change failure rate could indicate a problem with team skills, deployment process, or understanding and management of the deployment infrastructure.
- **Change Volume:** Number of new features or functions deployed in a period. A high change volume with a low failure rate suggests a high tempo of successful development. A high change volume with a high failure rate might indicate issues with the DevSecOps.
- **Issue Resolution Time:** Average time needed to resolve a reported issue. This measures the time it takes to identify and fix a reported defect or issue.
- **Issue Volume:** Number of issues reported in a given period, e.g., help desk ticket creation rate. A high issue volume might indicate customer dissatisfaction or production issues.
- **Mean Time To Recovery (MTTR):** Time between a failed deployment and full restoration of production operations. Short MTTR indicates a capable DevSecOps process; longer MTTR suggests problems.
- **Time to Patch:** Time between identifying a vulnerability in the application and successful production patch deployment. This is an indication of the ability of DevSecOps teams to find and fix defects.
- **Time to Value:** Time between a feature or function request and the realization of business value. Most businesses want to minimize time to value.

People

Context Tag:	CMMI-PPL
Context:	Use processes to align workforce performance and business objectives.

Identify and define measures in support of workforce management measurement and performance objectives. Consider the following types of measures:

- Cost per hire
- Number of initial candidates that transition beyond the probation period
- Demographics of candidates and hires, including factors such as source, age, diversity, equity, and inclusion
- Success rate and time spent on recruiting, selection, onboarding, and transitioning
- Time from opening a position to filling it
- Percent of unit members involved in personnel activities
- Rate of transitioning individuals into new positions
- Number of people undergoing outplacement

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

When selecting measures for managing service capacity and availability, consider supported activities, reporting requirements, information use. Commitments and agreements should include the selected measures and analytic techniques. Analysis should include describing the relationship between identified measures and requirements and deriving objectives that state specific target measures or ranges to meet for each measured attribute. Capacity and availability measures should be traced to requirements. Tools needed to support the collection and analysis of capacity and availability data should be identified, used, and kept updated.

Examples of availability measures include:

- Percentage available within agreed hours (this availability can be overall availability or component availability)
- Percentage unavailable within agreed hours (this unavailability can be overall unavailability or component unavailability)
- Duration of downtime due to failure, typically minutes, hours, or hours per week
- Failure frequency
- Degree of effect, e.g., number of affected users, number of minutes that users lost productivity, number of transactions or vital business functions not processed or carried out, number of applications impeded
- Response time of the system to incidents, transaction response times, and response times (as a capacity measure or availability measure)

- Reliability, e.g., number of breaks, mean time between failures, mean time between incidents

Examples of capacity and availability measures include:

- Use of limited resources
- Use of components
- Unused limited resources
- Unused components
- Throughput, e.g., number of concurrent users, number of transactions to process
- Queue length (maximum and average)
- Number of a type of resource or one or more specific resources in use a selected number of times (this use can be monitored by calendar time)

[Table MPM-5: Example Service Measurement Relationships](#) depicts some common examples of measurement relationships in the context of services.

Table MPM-5: Example Service Measurement Relationships

Example Business Objective	Information Need	Purpose	Information Type or Category	Example Base Measures	Example Derived Measures
Provide agreed service continuity.	Can the service provider recover services from disasters or major disruptions within agreed timeframes?	Provide insight into whether the service provider will implement service continuity plans successfully to provide agreed service continuity.	Service continuity	• Number of services with recovery test failures • Total number of services in the service catalog	• Service continuity confidence rate
Provide appropriate capacity to meet business needs.	Are there enough resources (or too many) to meet demand for services?	Provide insight into resource utilization, idle resources, and inadequate capacity to	Capacity	• Total number of service requests • Available service provider personnel hours • Service time	• Average service time • Service provider resource utilization

Example Business Objective	Information Need	Purpose	Information Type or Category	Example Base Measures	Example Derived Measures
		meet demand.			
Provide effective services.	How effective is the service?	Provide insight into percentage of reworked service requests.	Service performance	• Number of service requests reworked • Total number of service requests	• Service rework rate
Provide appropriate, agreed service availability.	Is the service provider providing appropriate, agreed service availability?	Provide insight into the availability of the service.	Availability	• Agreed service time • Downtime	• Availability
	Is the service reliable as agreed?	Provide insight into the reliability of the service.	Availability	• Available time (in hours) • Total downtime (in hours) • Number of breaks in service (interruptions to normal service)	• Reliability as mean time between failures (MTBF)

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

The acquirer uses defined supplier measures and measures of acquirer progress and output to manage the project. Define measures needed to manage the acquirer's performance objectives, some of which need to be provided by suppliers. Incorporate the measures that suppliers collect and report in the supplier agreement and service level agreements.

Define acceptance criteria to enable the intended use of supplier measures, such as potential aggregation and analysis. These criteria should include criteria associated with the collection and transfer mechanisms and procedures that the supplier performs. Consider all supplier measure characteristics that can affect their use, such as differences in financial calendars used by different suppliers.

The supplier may provide measures as detailed measurement data or measurement reports. Align measures that come from suppliers with the acquirer's acceptance criteria for supplier measures. Record acceptance criteria in measurement specifications or checklists.

MPM 2.3

Required Practice Information

Practice Statement

Obtain specified measurement data according to the operational definitions.

Value

Improves decisions and increases the likelihood of successfully completing projects.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Measurement results can help to:

- Monitor progress and performance
- Monitor response times
- Fulfill recorded obligations
- Make informed management and technical decisions
- Determine when corrective actions need to be taken
- Build historical data for use in future analysis

Example Activities

Example Activities	Further Explanation
Collect data for currently specified base measures.	
Calculate derived measures.	
Check data integrity as close to the source of data as possible.	Data integrity covers three aspects: • Accuracy and precision • Coverage • Completeness Checklists are useful for verifying data integrity.

Example Work Products

Example Work Products	Further Explanation
Base and derived measurement data	
Results of data integrity checks	

Context Specific

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

Obtain derived measures from the supplier as defined in the supplier agreement. The acquirer should ensure the supplier agreement specifies supplier data requirements for frequency, accuracy, and consistency. Follow up with suppliers if data is not available or data integrity checks indicate potential data errors.

Use acceptance criteria to verify:

- The results of data integrity tests conducted by the supplier
- The integrity of the supplier data

Example supplier deliverables include:

- Base and derived supplier measurement data sets
- Results of data integrity tests of supplier measurement data
- Supplier data collection frequency
- Regulatory and statutory compliance data

MPM 2.4

Required Practice Information

Practice Statement

Analyze performance and measurement data according to the operational definitions.

Value

Provides insight into performance and actions needed to meet objectives.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Compare measured performance to measurement and performance objectives to help determine the organization's ability to meet objectives. Identify potential areas of improvement by analyzing performance shortfalls. Use this information to develop, evaluate, and prioritize proposed improvements, and identify additional areas for analysis.

These activities provide the foundation for understanding performance with a wide range of stakeholders. It is important to make performance visible within and beyond the project. This communication includes the benefits obtained from performance improvements and may also be a trigger for further performance improvements.

Example Activities

Example Activities	Further Explanation
Perform analyses, interpret results as planned, and draw conclusions.	Typically, the analyses are specified in the operational definition. Results of planned analyses can identify the need to: • Complete additional, unanticipated analyses • Reevaluate capacity, availability, reliability, and maintainability trends; it may be advisable to perform a failure trend analysis • Refine existing measures • Calculate additional derived measures • Collect data for additional base measures to properly complete the planned analysis
Record analysis results and any significant deviations.	
Review results with affected stakeholders.	Reviewing the results with stakeholders can prevent misunderstandings and lead to improvements in the data analysis and reporting. Ensure that results are: • Understandable • Easily interpretable • Clearly tied to identified information needs and objectives The analysis may not be clear to those who are not measurement experts. Communication may include: • How to interpret results based on the evaluation methods used • How results address information needs • How results may affect the project

Example Activities	Further Explanation
	Examples of actions taken to help others to understand results include: • Providing training on the use and understanding of measurement results • Discussing the results with stakeholders • Modifying the approach for communication • Providing background and explanation
Refine measurements and analysis techniques in the operational definitions.	Performing data analyses and preparing results can lead to improved future efforts based on lessons learned. These lessons learned can lead to: • Improved measurement specifications • Improved data collection procedures • Ideas for refining information needs and objectives

Example Work Products

Example Work Products	Further Explanation
Performance data analysis results	May include: • Alignment to objectives • Results and conclusions • Identified performance shortcomings • Improvement candidates • Recommended actions
Updated operational definitions	

Related Practice Areas

Process Management (PCM)

Context Specific

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

The supplier agreement should specify supplier requirements for analyzing measurement data, including the definition and examples of measures the supplier must provide to the acquirer.

Example measurement analysis activities acquirers can perform with suppliers include:

- Discussing results and preliminary conclusions with suppliers
- Coordinating additional analyses with suppliers
- Reviewing initial results related to supplier progress or output with suppliers and determining if revisions are appropriate based on their response
- Updating data acceptance criteria for supplier measures

- Reviewing service level performance

Consider validating data collected from a supplier through periodic audits of the supplier for acquirer required measures. This may be defined in the supplier agreement.

Review specified analyses and reports with suppliers and identify their commitment to support the analysis. Review recommendations they may provide related to the analysis of measurement data.

MPM 2.5

Required Practice Information

Practice Statement

Store measurement data, measurement specifications, and analysis results according to the operational definitions.

Value

Enables analysis of performance to improve the likelihood of repeating successes.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Storing measurement data and information:

- Enables multiple options for analysis now and in the future
- Enables its timely and cost-effective use as historical data and results
- Makes historical data and results readily available to stakeholders

Provide sufficient context for interpretation or application of:

- Measurement data
- Analysis techniques
- Analysis results

Stored information contains or refers to other information needed to understand and interpret the measures and to assess them for reasonableness and applicability, e.g., measurement specifications used on different projects when comparing across projects.

Projects can choose to store specific data and results in a project-specific repository. When data are shared across projects, the data can reside in the organization's measurement repository.

Example Activities

Example Activities	Further Explanation
Review data to ensure their quality.	
Store data according to data storage procedures.	
Make stored data available for use.	
Prevent stored information from being used inappropriately.	Control access to data and educate people on the appropriate use of data. Examples of inappropriate data use include: • Disclosure of information provided in confidence • Faulty interpretations based on incomplete, out-of-context, or otherwise misleading information • Measures used to improperly evaluate the performance of people • Questioning the integrity of individuals Observe legal and regulatory requirements as to who can store which information and who can access them.

Example Work Products

Example Work Products	Further Explanation
Stored data	May include: • Context for data • Sets of data that were collected • Analysis reports and presentations • Retention period for stored data • Data security

Context Specific

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

The acquirer protects measurement data provided by the supplier according to the supplier agreement. The supplier agreement might specify that the acquirer must restrict access to a supplier's measurement data.

Information stored includes data acceptance criteria for supplier data.

The supplier agreement specifies:

- Measurement data the supplier must provide to the acquirer
- The format the supplier should use for providing data to the acquirer
- How the supplier will collect and store measurement data, e.g., retention period of data

- How and how often the supplier will transfer data to the acquirer
- Who has access to data
- Regulatory requirements
- Provisions for handling proprietary data

Develop mechanisms to transfer measurement data and process guidance from the supplier to the acquirer. The acquirer can integrate data collection from a supplier with periodic monitoring and review of supplier activities. In the supplier agreement, the acquirer should specify the applicable standard report formats and tools the supplier will use for reporting.

Review data collection and storage procedures with suppliers throughout the life of the agreement. Update data collection and storage procedures, as needed, and obtain supplier commitment to collect and store measurement data and reference procedures in the supplier agreement.

MPM 2.6

Required Practice Information

Practice Statement

Take actions to address identified issues with meeting measurement and performance objectives.

Value

Enables the ability to meet performance objectives.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Obtain performance information and data at the appropriate level, e.g., business, projects, process. Use analysis results to ensure sustainment and improvement. If the observed performance deviates from expected results, take appropriate actions to correct the deviation. Potential actions to correct significant deviations include:

- Performing causal analysis activities to identify and correct the cause of the deviation
- Replanning
- Initiating improvement activities
- Renegotiating objectives

Example Activities

Example Activities	Further Explanation
Record and implement corrective actions and manage to closure.	
Record and submit proposed improvements and communicate results.	

Example Work Products

Example Work Products	Further Explanation
Revised objectives, plans, and commitments	
Records of performance evaluations	May include: • Comparison of planned vs. actual performance • Proposed improvements
Records of significant deviations	May include: • Past actions and improvements • Side effects of actions and improvements
Proposed improvements	Include: • Source of the proposed improvements • What is improved by each proposed improvement

Related Practice Areas

Causal Analysis and Resolution (CAR)

Monitor and Control (MC)

Process Management (PCM)

Level 3

MPM 3.1

Required Practice Information

Practice Statement

Develop, keep updated, and use the organization's measurement and performance objectives traceable to business objectives.

Value

Optimizes resource usage to improve business success.

Additional Required Information

Develop, review, and analyze business objectives. Derive business objectives based on organizational structure, e.g., divisions, sectors, departments, services, projects, teams. The scale and variation of the allocation of objectives across sub-levels can depend on the business strategy and tactics, customer base, size, complexity, and solution lifecycle. Once completed at the organizational level, allocate the organization's measurement and performance objectives to the project level.

Explanatory Practice Information

Additional Explanatory Information

In order to define the organization's measurement and performance objectives, affected stakeholders should understand the:

- Organization's business model
- Business context
- Business objectives
- Related critical factors necessary to ensure the future success of the organization

These factors help to align measurement and performance needs and objectives at both the organizational and project levels.

Allocate the organization's measurement and performance objectives to appropriate subordinate units down to the project level. Review these allocated objectives, for meaning and usefulness within their context, with the affected stakeholders. Ensure that the measures and performance data used contribute to understanding performance at both the project and organizational level. Elevate project measurement and performance objectives as appropriate to the organizational level for deployment to other projects. Iterate this process between the organization and projects.

Example Activities

Example Activities	Further Explanation
Develop, review, and analyze the business objectives that drive measurement and performance objectives.	Analysis may include: • Business objectives • Current performance data and baselines • Current distribution of objectives to sub-units • Business and organizational models • Critical success factors • Customer data • Competitor data
Develop, record, and use the organization's measurement and performance objectives and keep them updated.	
Analyze the performance of projects within the organization.	Analyze this information periodically or on an event-driven basis to meet the organization's needs and to determine how it contributes to organizational performance and meeting the objectives.
Work with affected stakeholders to allocate the organization's measurement and performance objectives to projects.	
Trace new and revised measurement and performance objectives to business objectives.	
Review and update the allocations of measurement and performance objectives and communicate with affected stakeholders.	

Example Work Products

Example Work Products	Further Explanation
Business objectives	
Measurement objectives	Measurement objectives may address more than just performance, such as: • Measurement calibration data, e.g., accuracy, precision • Measurements for regulatory purposes or customer demands • Deployment, progress, and implementation status
Performance objectives	

Example Work Products	Further Explanation
Allocated measurement and performance objectives	Include: • Allocation of organization objectives to projects • Rationale and context for the allocation

MPM 3.2

Required Practice Information

Practice Statement

Follow organizational processes and standards to develop and use operational definitions for measures and keep them updated.

Value

Enables consistent collection, understanding, and use of organizational measurement and performance data to improve performance and increase likelihood of success.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Organizational processes and standards levy requirements for operational definitions that are to be implemented and used throughout the organization. This promotes consistent:

- Definition
- Collection and storage
- Aggregation
 - Across projects
 - Up to the organization or business level
- Analysis and understanding
- Reporting and recording

Example Activities

Example Activities	Further Explanation
Record, communicate, and use organizational standard operational definitions for selected measures and keep them updated.	
Revise the set of operational definitions of measures as needed.	Periodically evaluate measures for their continued usefulness.

Example Work Products

Example Work Products	Further Explanation
Operational definitions following organizational standards	

MPM 3.3

Required Practice Information

Practice Statement

Develop, keep updated, and follow a data quality process.

Value

Ensures that use of the measurement and performance data results in better decision-making.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Data quality focuses on integrity in areas such as the completeness and correctness of performance and measurement information. Data quality provides a foundation to verify:

- Measures adhere to their operational definitions
- Data integrity of the measurement repository enables:
 - Efficient and effective operations, decision-making, and planning
 - Decision error reduction
 - Performance improvement

Develop a process to analyze and improve data quality by:

- Minimizing measurement system errors
- Introducing controls to ensure that data inputs are valid
- Ensuring that metrics and measurements support effective decision-making
- Understanding the accuracy, completeness, and coverage of the data
- Providing adequate training

Example Activities

Example Activities	Further Explanation
Develop criteria for data quality, accuracy, precision, and validity.	

Example Activities	Further Explanation
Objectively evaluate the measurement repository and data to determine if the data quality criteria are met.	Determine and plan the frequency of evaluations.
Identify and communicate data quality issues and track to closure.	
Identify and communicate data improvement proposals.	

Example Work Products

Example Work Products	Further Explanation
Data quality criteria	
Measurement repository data quality report	
Data quality issues	
Improvement proposals	

MPM 3.4

Required Practice Information

Practice Statement

Develop, keep updated, and use the organization's measurement repository.

Value

Supports informed decisions leading to more successful projects through timely access to measurement and performance data.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Develop a repository to store and retrieve measurement data that can be used to:

- Understand and improve performance
- Support the evolution of a more efficient and effective data set for use throughout the business
- Improve and sustain processes
- Support planning

- Support analysis

The measurement repository:

- Contains project, process, and performance measures that are related to the organization's set of standard processes.
- Contains or refers to the information needed to understand, interpret, and assess measures and performance for reasonableness and applicability.
- Contains up-to-date and correct information. Maintaining the repository is important since objectives and related measures and performance change over time.

Example Activities

Example Activities	Further Explanation
Determine the organization's needs for storing, retrieving, and analyzing measurements.	This results in understanding and recording the requirements for the measurement repository.
Design and implement the measurement repository.	Functions of the measurement repository include: • Supporting effective comparison and interpretation of measurement and performance data • Providing sufficient context to quickly identify and access data in the repository • Improving the estimation, measurement, and performance accuracy and precision through use of historical data • Supporting the understanding of performance
Populate the contents and communicate the availability and benefits of the measurement repository.	
Revise the measurement repository as needs change.	For example, revise when: • Measurement and performance objectives change • New processes are added • Processes are revised and new measures are needed • Data with finer granularity is required • Greater visibility into the process and performance is required • Measures are no longer used

Example Work Products

Example Work Products	Further Explanation
Measurement repository design	The system design should address: • Repository structure • Required content • Related systems or subsystems

Example Work Products	Further Explanation
	• Collection • Support environment • Retrieval mechanisms • Security • Retention • Maintenance • Data validation • Standard reporting
Organization's measurement repository	
Organization's measurement data	

MPM 3.5

Required Practice Information

Practice Statement

Analyze organizational performance using measurement and performance data to determine and address performance improvement needs.

Value

Contributes to business success through the analysis and improvement of performance.

Additional Required Information

Identify, record, and use aggregation rules for measurement and performance data and keep them updated.

Explanatory Practice Information

Additional Explanatory Information

Identify potential areas for improvement through analysis at the organizational level. Determine areas that could address process performance shortfalls. Use analysis to evaluate and prioritize potential improvements. Identify processes and technologies that will have the largest effect on achieving those improvements. Ensure that each performance improvement is traceable to processes and business objectives.

Compare data before and after performance improvements are implemented to ensure the improvements were effective. Aggregate data from various projects to the organizational level and then compare the results with business objectives. Communicate the results, achieved benefits, and the satisfaction of objectives to stakeholders.

Example Activities

Example Activities	Further Explanation
Aggregate performance data to the organizational level.	
Analyze measurement and performance objectives against current performance to evaluate the organization's ability to satisfy business objectives.	May include: • Which processes are involved in and contribute to a selected performance improvement • An estimate of the contribution of the respective processes • Resources consumed during operations, delivery, and use • Traced contribution of performance improvements to related objectives • Any barriers that may hinder meeting objectives or successful deployment of improvements Review and obtain agreement with affected stakeholders for service level targets.
Develop, use, and keep updated descriptions of the resources consumed, as well as planned and current service system performance.	Examples include: • Identify flow charts for the service system and its processes • Record of the system's current capacity and availability, which can require determining the capacity and availability of service system components • Consider service system graphical representations from collected measurements and analyses ○ Estimate the capacity and availability of the service system at peak work volumes ○ Develop capacity and availability models and keep them updated. Common models depict demand data, capacity and availability data, availability data, performance data, representations of resource use, representations of resource needs, data on the use of resources ○ Define thresholds, typically below the level at which an exception condition or breach of requirement occurs to allow time for corrective action, and associated with demand, workload, use of resources, and performance to define exception conditions in the service system and breaches or near breaches of requirements ○ Identify thresholds that define exception conditions and breaches
Identify shortfalls and potential improvement areas where actual performance is not meeting business objectives.	Performance shortfalls may include not meeting objectives for: • Productivity • Cycle time • Customer satisfaction

Example Activities	Further Explanation
	Examples of areas to consider for improvement include: • Product technology • Process technology • Personnel development • Team and organizational structures • Infrastructure • Supplier selection and management • Personnel development, including individual, workgroup, and organizational competency improvements
Develop, use, and keep updated service system representations from collected measurements and analyses.	
Record performance improvement needs.	Use this information and evaluation results when developing or updating proposed improvements for performance.
Analyze and record anticipated cost and benefits associated with addressing performance improvement needs.	May include: • Resources including schedule, effort, people, processes, tools, etc. • Barriers that may hinder successful deployment • Detailed organizational changes needed to implement each improvement • Verification and validation activities
Communicate results to affected stakeholders.	
Submit performance improvement suggestions.	

Example Work Products

Example Work Products	Further Explanation
Aggregated performance results	Elicit performance and improvement results from the project level. Results may include: • Organization's measurement and performance objectives • Results of process and work benchmarking efforts • Measured effectiveness of work environments • Analysis of work and organizational performance compared to quality and productivity objectives, including: ○ Aggregation to the business level ○ Effect on satisfying business objectives

Example Work Products	Further Explanation
	Effect on meeting measurement and performance objectives
Performance analysis results	May include: - Identified processes or technologies that critically contribute to performance improvements - Rationale for why identified processes or technologies are critical - Identified shortfalls, e.g., risks, issues, or achievements, towards meeting measurement and performance objectives at the project and organizational level
Performance improvement needs	Includes traceability of improvements to processes and business objectives.
Submitted performance improvement suggestions	

Related Practice Areas

Process Management (PCM)

Context Specific

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

When determining performance needs for services, develop, keep updated, and use models, thresholds, and targets for supporting capacity and availability management to keep the work on track to meet objectives.

Capacity and availability management may include developing service system representations used for delivering the solution and using these representations for:

- Supporting negotiation of appropriate agreements
- Planning
- Making decisions
- Considering corrective actions
- Providing and allocating resources to meet current and future requirements

These representations provide insight into the behavior of a service system given specific work volumes and varieties. These representations can be built using diagrams; spreadsheets; commercial off-the-shelf (COTS) tools, e.g., simulation packages; prototypes; or tools developed in house. For some service delivery systems, the representations are historical baselines; trend analyses; analytical models; analysis of waiting times in queues; simulation models; statistical models, e.g., regression models, time series models; causal models, e.g., probabilistic networks; or application sizing.

MPM 3.6

Required Practice Information

Practice Statement

Periodically communicate performance results to the organization.

Value

Enhances coordination and understanding of performance and improvement value to reduce waste and increase the likelihood of achieving objectives.

Additional Required Information

Performance results include impact, benefits, and analysis of performance against objectives.

Explanatory Practice Information

Additional Explanatory Information

Communicate composite measurement and performance results. Widespread communication leads to greater understanding across the organization of the benefits of performance improvement. This understanding is key to developing a sustainable improvement culture with a positive attitude towards measurement and performance management and process improvement.

Example Activities

Example Activities	Further Explanation
Develop and record performance improvement reports.	
Communicate performance improvement results to affected stakeholders.	

Example Work Products

Example Work Product	Further Explanation
Performance improvement and analysis reports	May include: • Contextual information or guidance to help interpret analysis results • Discussion and interpretation of results • Usage of results • Performance improvement results from projects • Effect on satisfying measurement and performance objectives • Aggregation to the business level • Effect on satisfying business objectives

Level 4

MPM 4.1

Required Practice Information

Practice Statement

Use statistical and other quantitative techniques to develop, keep updated, and communicate quality and process performance objectives that are traceable to business objectives.

Value

Establishes realistic quality and process performance objectives enabling better decision-making, increasing the likelihood of meeting business objectives.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Use statistical techniques to determine Quality and Process Performance Objectives (QPPOs) for the work. QPPOs can be used at the project, organizational level, or any level. Carefully select performance data for statistical analysis as it can be expensive. Analyze information periodically and on an event-driven basis. Trace QPPOs to business objectives.

Negotiate the QPPOs for the work with enough detail to allow for evaluating the objectives and risks. The objectives may be expressed as either a distribution or discrete number. Update them:

- As the project's understanding of actual performance and variation becomes known and more predictable
- To reflect the changing needs and priorities of the business
- When higher level objectives are changed

QPPOs may address:

- Planned objectives to be achieved resulting from improvements
- Current objectives to be sustained

Example Activities

Example Activities	Further Explanation
Define, record, keep updated, and communicate Quality and Process Performance Objectives QPPOs.	This involves: • Incorporating appropriate business and organizational objectives

Example Activities	Further Explanation
	• Recording objectives that reflect the quality and process performance needs and priorities of affected stakeholders • Determining how each objective will be achieved One common technique for recording these objectives is the SMART approach: • Specific • Measurable • Achievable • Relevant • Time-bound Ensure traceability between the performance and business objectives. Techniques that aid in defining QPPOs may include: • Use of control charts • Analysis of variation • Regression analysis • Use of confidence or prediction intervals • Sensitivity analysis • Simulations • Hypothesis tests
Derive interim objectives to monitor progress toward achieving the stated objectives.	Establish interim objectives for characteristics of selected phases, milestones, work products, and processes.
Determine and record the risk of not achieving QPPOs.	
Resolve conflicts among the QPPOs, e.g., if one objective cannot be achieved without compromising another.	To resolve conflicts: • Set relative priorities for objectives • Consider alternative objectives in light of long-term business strategies as well as short-term needs • Involve and negotiate with affected stakeholders in tradeoff decisions Additionally, process performance baselines and models can help identify suboptimized processes or subprocesses that may conflict with other QPPOs. Revise objectives, as necessary, to reflect results of conflict resolution.

Example Work Products

Example Work Products	Further Explanation
QPPOs	Examples of project QPPOs include:

Example Work Products	Further Explanation
	• Maintain change request volume below a target value • Reduce idle time by x% by a target date • Maintain schedule slippage below a specified percentage Examples of measurable quality characteristics include: • Performance • Defects or issues • Critical resource utilization • Severity of customer complaints Examples of measurable process performance characteristics include: • Cycle time • Percentage of rework time • Defect escape rates
Risks of not achieving the QPPOs	

Related Practice Areas

Risk and Opportunity Management (RSK)

Context Specific

Data

Context Tag:	CMMI-DATA
Context:	Use processes to incorporate data management best practices as an integral part of the solution.

Examples of Quality and Process Performance Objectives (QPPOs) in consideration of managing data include:

- Data completeness, e.g., 98% of expected data values are captured
- Data accuracy, e.g., less than 5% data defect rate found during data verification
- Data coverage, e.g., 100% coverage across all mission critical datasets
- Data processing time, e.g., 90% of data requests processed within 24 hours

Development

Context Tag:	CMMI-DEV
Context:	Use processes to develop quality products and services to meet the needs of customers and end users.

Examples of Quality and Process Performance Objectives (QPPOs) using derived targets include:

- Maintain a code review rate between 75 to 100 lines of code per hour

- Keep test rate over a specified number of test cases per day
- Maintain productivity in generating use cases per day
- Keep design complexity (fan-out rate) below a specified threshold

MPM 4.2

Required Practice Information

Practice Statement

Select measures and analytic techniques to quantitatively manage performance to achieve quality and process performance objectives.

Value

Focuses measurement and management activities on the data that provide the most insight into achieving the objectives.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Measurement and analytic techniques enable preparation for statistical and quantitative management of performance, by establishing a traceable relationship of the selected measures, and their analyses, to the objectives. Use appropriate analytic techniques, including data visualization, to enable users to recognize significant deviations from performance objectives and take corrective action.

Example Activities

Example Activities	Further Explanation
Identify common measures from the organizational measurement repositories.	
Identify additional measures that may be needed to cover the critical work product and process attributes of the selected processes.	
Select measures to manage processes using statistical and other quantitative techniques.	Selection should not be limited to solution, progress, or performance measures only. Measures can be used to develop analysis, process, and success indicators which provide better insight into process performance.

Example Activities	Further Explanation
	Measures that aggregate data from multiple sources or over time typically mask underlying problems, making problem identification and resolution difficult. For short projects, it may be necessary to aggregate data across similar instances of a process to enable analysis of its process performance.
Specify the operational definitions of new measures.	
Identify the statistical and other quantitative techniques to be used.	These statistical and other quantitative techniques help to: • Characterize process variation • Recognize when variation is excessive, and investigate why • Identify when statistically unexpected behavior occurs Examples of statistical techniques to analyze process performance include: • Statistical process control • Regression analysis • Analysis of variance including central tendency and distribution • Time series analysis • Hypothesis testing for statistical significance Graphical displays can help: • Visualize process performance • Understand relationships between two or more sets of data • Track variation over time • Identify problems or opportunities • Evaluate the effects of different factors Examples of graphical displays include: • Scatterplots • Control Charts • Histograms • Box and whisker plots • Run charts • Ishikawa diagrams
Analyze the relationship of identified measures to the Quality and Process Performance Objectives (QPPOs) to derive targets.	Examples of QPPOs using derived targets include: • Keep requirements gathering sessions to under three hours • Maintain rework levels below a specified percent • Maintain productivity in generating a number of work products per day
Develop the environment to support collection, derivation, and analysis of	This implementation is based on the: • Description of the organization's set of standard processes • Description of the defined process in use • Capabilities of the support environment

Example Activities	Further Explanation
new measures and keep it updated.	
Record, keep updated, and use measures and statistical analysis techniques.	

Example Work Products

Example Work Products	Further Explanation
List of selected and new measures	Include: • Operational definition of new measures suitable to support statistical and other quantitative management • Identified (or linked to) statistical and other quantitative techniques to analyze the measures • Representations of data and analysis results
Repository of analytical and statistical techniques	Include: • Definition of analytical and statistical techniques • Links to measures • Needed skills for applying the defined techniques • Identified people who have the needed skills • Tools to support the techniques
Environment to support collection, derivation, and analysis of new measures	
Results of analysis and their derived targets	
Traceability to objectives	Show the relationship between the measures, the QPPOs, and higher-level business objectives.

MPM 4.3

Required Practice Information

Practice Statement

Use statistical and other quantitative techniques to develop and analyze process performance baselines and keep them updated.

Value

Enables quantitative understanding of performance and capability to ensure that objectives can be met.

Additional Required Information

A process performance baseline is a recorded, statistical characterization of the actual results achieved. Derive baselines by analyzing measures to identify central tendency and distribution or range of results that characterize the expected performance. Use these baselines to determine the expected results of the process under a given set of circumstances. The organization's process performance baselines measure performance for selected processes within the organization's set of standard processes.

Baselines provide an understanding of process variation and support informed decision-making through a better understanding of risks to achieving objectives. Baselines enable:

- Determining the stability and capability of the process
- Detecting defects earlier to save resources
- Detecting significant cost or schedule issues earlier to take corrective actions
- Facilitating earlier detection of anomalies with critical processes

A stable process is important to creating reliable process performance baselines. Understanding the presence (or absence) of variation enables more accurate understanding of the process capability. In determining stability and capability of the process, two key terms come into play:

- **Stable process:** The state in which special causes of process variation have been removed from the process and prevented from recurring. In a stable process, only common cause variation of the process remains.
- **Capable process:** A stable process that can meet the Quality and Process Performance Objectives (QPPOs) set for it, and the variation of the process falls within set specification limits.

Explanatory Practice Information

Additional Explanatory Information

Process performance baselines may be used to:

- Compare to performance objectives to identify if they are being met
- Establish trial bounds when starting a control chart
- Compare actual process performance among multiple projects
- Establish benchmarks
- Enable identification of special cause of variation, triggering potential root cause analysis

The work that the process performance baselines address includes:

- Individual process activities within a larger process
- Sequences of connected activities
- Processes whose activities are performed by different workgroups
- Activities that cover the entire lifecycle
- Activities for developing individual work products
- Determining inherent process variation, i.e., common cause of variation, in performance
- Determining assignable process variation, i.e., special cause of variation, in performance

- Understanding the impact of variation on stability and capability on the performance of the process and achievement of the QPPOs
- Understanding how capable a process is in meeting its QPPOs

There can be several performance baselines to characterize performance for subgroups. Examples of subgroup categories include:

- Line of business
- Domain or function
- Complexity
- Project size
- Work product size
- Process elements or attributes from the organization's set of standard processes

Tailoring the organization's set of standard processes can significantly affect the comparability of data and how data is included in process performance baselines.

Conditions that may affect the performance of processes and justify the development of separate process performance baselines include:

- The level of experience or proficiency of the individuals performing the processes
- Organizational or business conditions
- Specific methods or tools used in performing the processes
- The nature of the solution for which the processes are performed

Statistics used to characterize process performance baselines include:

- Expected performance as measured by mean, median, mode, or other measures of central tendency
- Performance variation as measured by upper and lower control limits, standard deviation, interquartile range, range, or other measures of variation
- Shape of the distribution as measured by statistics, e.g., skewness, kurtosis
- How performance parameters may vary under different conditions
- Performance trends over time

Example Activities

Example Activities	Further Explanation
Analyze the collected measures to establish central tendency and distribution or range of results that characterize the expected performance of selected processes.	Include the stability and capability (when specification limits are defined). This analysis can help decide the best selection or subset of process performance baselines to maintain. Ensure that the baselines are linked to QPPOs and business objectives.
Record, keep updated, and use process performance baselines.	Examples may include: • Statistical process control charts • Box and whisker diagrams

Example Activities	Further Explanation
	• Confidence intervals
Review process performance baselines with affected stakeholders.	Stakeholders can help determine if the baselines are appropriate.
Make the process performance baselines available in the organization's measurement repository and communicate to the organization.	Workgroups use the organization's process performance baselines to estimate the natural bounds for their process performance. Some project process performance baselines may not be included in the organization's measurement repository.
Revise the process performance baselines as needed.	For example, when: • Processes change • The organization's results change • The organization's needs change • Suppliers' processes change • Suppliers change

Example Work Products

Example Work Products	Further Explanation
Results of analysis of process performance data	
Process performance baselines	Include: • Central tendency • Range and distribution • Description of the context for the data • Reference to the operational definition of the data to enable accurate interpretation • Links to QPPOs

MPM 4.4

Required Practice Information

Practice Statement

Use statistical and other quantitative techniques to develop and analyze process performance models and keep them updated.

Value

Reduces cost and increases quality by predicting likelihood of meeting objectives and allowing for early corrective action.

Additional Required Information

A process performance model:

- Developed from historical process performance data such as those contained in process performance baselines
- Describes, models, or depicts variation in measurable attribute values and terms
- Predicts interim or final process performance
- Estimates expected range and variation of predicted results
- Includes at least one measurable attribute representing a controllable input tied to a subprocess

The condition in the last bullet makes what-if analyses possible: controllable input(s) can be varied and resulting changes in process performance predicted. Perform analyses during planning, dynamic replanning, and problem resolution to make process tailoring choices that maximize likelihood of meeting Quality and Process Performance Objectives (QPPOs).

Process performance models can be:

- Statistical, e.g., regression
- Probabilistic, e.g., Bayesian
- Simulation-based, e.g., Monte Carlo or discrete event simulation

A process performance model can be a collection of models that when combined meet the criteria of a process performance model.

Explanatory Practice Information

Additional Explanatory Information

High Maturity organizations generally develop and keep updated a set of process performance models at various levels of detail to predict process performance. These models cover a range of activities and work product measures that are common across the organization and address the likelihood of achieving the organization's QPPOs.

Process performance models can help to:

- Predict performance results including confidence limits of selected processes
- Analyze and predict the performance associated with processes and changes to the organization's set of standard processes
- Assess return on investment for process and performance improvement activities
- Select processes that give the projects the highest probability of success
- Enable "what if" analysis for potential changes and improvements
- Estimate progress toward achieving the QPPOs

Examples of performance models include:

- Regression
- Complexity
- Discrete event simulation
- Monte Carlo simulation

Example Activities

Example Activities	Further Explanation
Establish process performance models.	Examples of analyses used to develop performance models include: • Visual analysis techniques, e.g., scatter plot • Regression or multivariate predictive techniques • Stochastic techniques • Classification analyses, e.g., defect or problem types • Monte Carlo analysis
Validate the process performance model.	One typical method of validation is to use historical information, e.g., use the first 9 months of data to predict the last 3 and then compare to actual results.
Calibrate process performance models based on the results.	
Review process performance models with affected stakeholders.	Stakeholders can help determine if the models are appropriate.
Make the process performance models available in the organization's measurement repository and communicate to the organization.	Some project process performance models may not be included in the organization's measurement repository.
Support the project's use of process performance models.	
Revise process performance models as needed.	For example, when: • Processes change • The organization's results change • The organization's measurement and performance objectives change • The organization's business or business objectives change

Example Work Products

Example Work Products	Further Explanation
Process performance models	May include: • Guidance for use • Description of model: ○ Equation or scenario ○ Controllable factors ○ Uncontrollable factors ○ Confidence and prediction limits
Validation results	Include results from utilizing the models.

Example Work Products	Further Explanation
Calibration results	

MPM 4.5

Required Practice Information

Practice Statement

Use statistical and other quantitative techniques to determine or predict achievement of quality and process performance objectives.

Value

Facilitates a quantitative understanding of risks to achieving objectives which maximizes likelihood of success.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Use statistical and other quantitative techniques to:

- Analyze variation in process performance
- Monitor the selected processes that drive achieving the objectives
- Review the Quality and Process Performance Objectives (QPPOs) to determine:
 - Their relationship to the business objectives
 - The risks associated with not meeting them
 - The actions needed to achieve them

Use multiple inputs to predict if the QPPOs will be satisfied. Quantitative models of performance can range from simple descriptive statistics concerning capability to sophisticated Bayesian, stochastic, or multivariate predictive models. These models may be used to predict project, team, or organizational performance based on the current capability of processes and the conditions that affect them. An organization may begin with standard models from related industries and over time refine their algorithms or parameters with internal data and experience. Quantitative models may differ in:

- Purpose
- Sophistication
- Analytic foundation
- Parameters
- Predictability
- Use among the various processes

Examples of quantitative models include:

- Models to predict the likelihood of achieving QPPOs from the aggregated performance results of projects
- Predictive models of characteristics most likely to affect capability and performance of the project and used to select improvements and adjust performance expectations
- Models of the effect of variations in practices and activities on the resulting capability and performance of processes
- Models for evaluating decisions involving performance tradeoffs

Example Activities

Example Activities	Further Explanation
Analyze the variation and stability of the selected processes and address deficiencies.	This analysis may involve: • Evaluating measurements in relation to the natural bounds and specification limits • Identifying outliers or other signals of potential non-random behavior (applying run rules or tests to identify anomalies) • Determining causes of outliers • Preventing or mitigating the effects of outlier recurrence, e.g., addressing special causes of variation Consider: • The sufficiency and accuracy of the data • Shifts in process performance that can affect the ability to achieve or maintain process stability • Deficiencies in process performance such as when there is too much variation to achieve the objectives Confirm that the process performance is stable before determining capability. Addressing stability typically includes an in-depth understanding of special causes. Addressing capability typically includes an in-depth understanding of common causes and addressing them appropriately. Analytic techniques for identifying outliers or signals include: • Statistical process control charts • Prediction, confidence, or tolerance intervals • Analysis of variation
Implement actions needed to address deficiencies in achieving QPPOs.	Examples of actions to address deficiencies in achieving objectives may include: • Improving the implementation of the existing process to reduce its variation or improve its performance, e.g., addressing common causes of variation or changing the process • Adopting new processes and technologies • Identifying risks and risk mitigation strategies for deficiencies

Example Activities	Further Explanation
	Renegotiating QPPOs when the processes are not capable of achieving them Some actions involve the use of causal analysis. Corrective actions may result in changes to attributes or measures related to controllable factors in process performance models. Process performance models can then be used to predict the effects of the actions. When taking critical corrective actions in high-risk situations, a process performance model can be developed to predict the effects of the change. Processes not selected for their impact on objectives can still cause problems or risks. Some level of monitoring for these processes may be helpful.
Use validated process performance models calibrated with data to assess progress toward achieving QPPOs.	For example, use process performance models to predict the latent defects in work products in future phases or in the delivered solution.
Identify and manage risks associated with achieving QPPOs.	
Record and communicate the results of the analysis, decisions, and actions identified.	

Example Work Products

Example Work Products	Further Explanation
Results of analysis, validation, and calibration	May include: - Graphs, charts, and data tables that support quantitative management - The range of process performance for each selected process attribute
Predictions of results to be achieved relative to the QPPOs	
Recorded risks of not achieving the QPPOs	
List of actions needed to address deficiencies in the process stability or capability of each selected process	

Related Practice Areas

Causal Analysis and Resolution (CAR)

Risk and Opportunity Management (RSK)

Level 5

MPM 5.1

Required Practice Information

Practice Statement

Use statistical and other quantitative techniques to ensure that business objectives are aligned with business strategy to optimize performance.

Value

Minimizes waste and rework through a more accurate understanding of capability which increases the likelihood of setting and meeting reasonable objectives.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Ensure alignment of Quality and Process Performance Objectives (QPPOs), business objectives, and business strategies.

Use organizational performance data, characterized by process performance baselines and models, to evaluate whether business objectives are realistic and aligned with business strategies and to understand variation. There may be multiple baselines and models needed in an organization to cover different types or aspects of work. After business objectives have been revised and prioritized, develop, keep updated, and communicate resulting QPPOs. Use process performance models to understand the processes and relationships needed to achieve the objectives and to perform what-if analyses to aid in the alignment process.

Example Activities

Example Activities	Further Explanation
Evaluate and update business objectives periodically and on an event driven basis to ensure that they align with business strategies.	Business objectives, business strategies, and organizational performance may change over time or become obsolete based on the needs of the organization or the strategies. Understanding performance in this context requires using statistical and quantitative techniques.
Compare business objectives with baselines and process performance predictions to ensure the objectives are realistic.	Business objectives can set the bar too high to motivate real improvement. Using process performance baselines and models helps balance expectations and reality. If process performance baselines and models are not available, use sampling techniques to quickly develop a quantitative basis for comparison.

Example Activities	Further Explanation
Prioritize business objectives based on recorded criteria.	Example criteria include the ability to: • Win new business • Retain existing clients • Respond to changing or disruptive markets • Innovate • Accomplish other key business strategies
Maintain QPPOs to address changes in business objectives.	Business and QPPOs typically evolve over time. Ensure existing objectives are met while identifying and managing new business objectives and associated QPPOs.
Revise measures to align with QPPOs.	
Communicate changes to business strategies, business objectives, and QPPOs to stakeholders.	

Example Work Products

Example Work Products	Further Explanation
Results of analysis of current performance against: • QPPOs • Business objectives • Business strategies	This is typically an iterative process that may be done from a bottom-up (QPPOs up to business strategies) or top-down approach, depending on the need and purpose.
Revised business objectives and strategies	
Revised QPPOs	
Revised measures	

MPM 5.2

Required Practice Information

Practice Statement

Analyze performance data using statistical and other quantitative techniques to determine the organization's ability to satisfy selected business objectives and identify potential areas for optimizing performance.

Value

Identifies areas that pose the greatest risk to achieving business objectives or greatest opportunity for increasing business performance.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Process performance baselines and models help to understand the current capability of the organization. Compare process performance baselines or predictions from process performance models against Quality and Process Performance Objectives (QPPOs) to determine the organization's ability to meet business objectives.

Use analysis to further refine the potential areas for improvement. This can be combined with Causal Analysis and Resolution practices to help diagnose and resolve root causes of the issues.

Example Activities

Example Activities	Further Explanation
Periodically and on an event-driven basis, compare QPPOs to current process performance baselines and models to evaluate and predict the organization's ability to satisfy business objectives.	For example, if cycle time is a critical business need, collect various measurements that support or influence characterizing cycle time. Compare overall cycle time performance data to business objectives to understand if the expected performance will satisfy business objectives.
Identify shortfalls where performance is not satisfying business objectives.	
Identify potential improvement areas based on the analysis of performance shortfalls.	Examples of areas to consider for improvement include: • Product technology • Process technology • Personnel development • Productivity • Team structures • Supplier selection and management • Other organizational infrastructures
Communicate results and develop a list of improvement proposals.	

Example Work Products

Example Work Products	Further Explanation
Performance analysis results	Include an analysis of current performance data versus business objectives.

Example Work Products	Further Explanation
List of performance shortfalls	Shortfall descriptions may include: • Context • Impact (including any interdependencies on other performance or processes) • Priority • Risks • Potential corrective actions
List of potential improvement proposals	

Related Practice Areas

Causal Analysis and Resolution (CAR)

MPM 5.3

Required Practice Information

Practice Statement

Select and implement improvement proposals based on the statistical and quantitative analysis of the expected effect of proposed improvements on meeting and optimizing business, quality, and process performance objectives.

Value

Increases likelihood that selected improvements will significantly contribute to achieving business, quality, and process performance objectives.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Proposed improvements selected for implementation should significantly contribute to achieving business, quality, and process performance objectives. Improvement ideas may come from either inside or outside the organization. By analyzing the benefits and impacts of improvements, the organization can prepare for their deployment and maximize the benefits.

Example Activities

Example Activities	Further Explanation
Use process performance models to predict process performance based on proposed improvements.	

Example Activities	Further Explanation
Run pilots to determine preliminary process performance based on proposed improvements.	
Run tests for statistical significance of modeled or piloted results.	
Analyze the costs and benefits of proposed improvements.	Estimate the cost, effort, and schedule required for implementing, verifying, and deploying each proposed improvement. Identify and use statistical and other quantitative data and techniques. Process performance models provide insight into the effect of process changes on process capability and performance. Criteria for evaluating costs and benefits may include: • Contribution toward meeting the organization's Quality and Process Performance Objectives (QPPOs) • Effect on mitigating identified work and organizational risks • Ability to respond quickly to changes in requirements, market situations, and the business environment • Impacts to commitments and skills • Effect on related processes, training, work environment, and technology • Cost of defining and collecting data that support the measurement and analysis of the process and technology improvement • Scalability • Expected life span of the improvement
Identify potential barriers and risks to deploying each proposed improvement.	Examples of barriers include: • Turf guarding and parochial barriers • Unclear or weak business rationale • Lack of short-term benefits and visible successes • Unclear picture of what is expected • Too many changes at the same time • Lack of involvement and support from affected stakeholders Examples of risks include: • Compatibility of the improvement with existing processes • Experience and skills of potential users • Complexity of the improvement • Difficulty implementing the improvement

Example Activities	Further Explanation
	• Ability to demonstrate the value of the improvement before widespread deployment • Justification for large, up-front investments in areas such as tools and training • Resistance to change
Record the evaluation results and decide to implement in accordance with evaluation criteria	The results may include: • Problem statement • Description of proposed improvements • Statistical and quantitative analysis of proposed improvements including evaluation criteria • Cost benefit analysis of proposed improvement • Implementation decision
Determine and record the validation method that will be used before broad-scale deployment of the change.	Determining the validation method includes defining the statistical or quantitative success criteria that will be used to evaluate results of the validation. Since innovations typically represent a major change to the process, most innovative improvements should be piloted. Other validation methods, including modeling and simulation, can also be used.
Implement selected proposals	
Validate implemented proposals before broad-scale deployment	Follow the validation methods recorded above
Deploy validated proposals	

Example Work Products

Example Work Products	Further Explanation
Analysis of potential impacts of proposed improvements	Analysis options include a: • Statistically significant prediction of likelihood of achieving the intended effect, e.g., from process performance models • What-if analysis
Piloting report	May include: • Results from piloting performance improvements against defined success criteria • Probability that the intended effect will be achieved when the improvement is deployed
Cost-benefit analysis results	May include: • Cost • Expected benefits • Organizational impact • Effect measured in quantitative and statistical terms

Example Work Products	Further Explanation
	• Relationship to and impact on business and QPPOs • Identification of proposed improvements
List of potential barriers and risks to implement the improvement	
Recorded validation methods	
List of submitted proposals for implementation	

Related Practice Areas

Process Asset Development (PAD)

Process Management (PCM)